

Hebrew Semantic Retrieval for Social-Rights Questions

Bridging everyday language and official phrasing with NLP

Author: Yanik Gotie (318553526) · Track 3 — Data-Driven NLP · HIT · Dr. Igor Kleiner

1. Introduction & Research Question

Israelis seeking social rights (benefits, allowances, bureaucratic procedures) ask questions in **everyday language** — "מה מגיע לי, מה פירוט אותי?" — while official sources are written in **formal, legal-administrative Hebrew** ("זכאות לדמי זכאות לבטלה"). This vocabulary gap makes simple keyword search miss relevant pages.

Research question. Does **semantic** retrieval with Hebrew sentence embeddings (AlephBERT) improve information finding in the rights domain over classic **lexical** retrieval (TF-IDF) — and does a **hybrid** of the two improve further?

Hypotheses. (H1) Semantic beats lexical on high *phrasing-gap* queries. (H2) A hybrid beats either single method on ranking-quality metrics (MRR, nDCG).

2. Data

A self-collected corpus (no Kaggle), from two public sources (details in `DATA.md`):

- **Kol-Zchut** (CC-BY-SA), via the official **MediaWiki API** — clean plain-text extracts across six life-domains: disability, unemployment, health, family, income-support, parenting.
- **Bituach Leumi** — a small curated set of benefit pages, scraped robots-aware from `#mainContent`.

Pages are cleaned (niqqud stripped, boilerplate removed) and **chunked** into ≤ 600 -char passages. **Snapshot: 296 documents → 1,602 passages → 122,928 tokens** (`data/processed/corpus.jsonl`).

Evaluation set. 40 hand-authored colloquial questions, each mapped to ground-truth document(s) and tagged with a `phrasing_gap` (low/medium/high). Scoring is at **document level**: a hit is a returned chunk whose parent document is in the gold set.

3. Methods

3.1 Classical NLP preprocessing

Each passage is run through a classical pipeline before lexical indexing:

- **Normalization** — strip niqqud (`[ - ]`), drop punctuation/Latin, lowercase, collapse spaces.
- **Tokenization** — whitespace/word tokens (`\b\w+\b`).
- **Stop-word removal** — a 94-word Hebrew function-word list (pronouns, prepositions, question words). Example: a 33-token passage → 26 content tokens after removing `על, כדי, וכן, מי, או`.
- **Vectorization (TF-IDF)** — 1-2-grams, `min_df=2`, sublinear TF, L2-normalized. The corpus becomes a **1,602 × 20,981** matrix (7,323 unigrams + 13,658 bigrams), **99.58 % sparse** (only 141,493 non-zeros). Top terms by document frequency: `נכות, עבודה, הכנסה, לאומי, לביטוח`, and the bigram `"לביטוח לאומי"` (264 docs).

- **Morphology note.** Hebrew glues particles (כ/ש/ו/ה/מ/ל/ב) to words, inflating the vocabulary. A light prefix-stripping experiment shrinks the raw vocabulary **13,415 → 8,963 types (-33 %)** — evidence that morphology, not topic, drives much of the sparsity. We kept full forms for the main results (no reliable offline Hebrew lemmatizer) and flag lemmatization as future work.

3.2 Retrievers (≥ 2 required → five built)

Model	Type	Idea
TF-IDF	lexical (baseline)	sparse cosine over the matrix above
BM25	lexical (bonus)	Okapi BM25 ranking
AlephBERT	semantic	imvladikon/sentence-transformers-alephbert, dense cosine
Hybrid-RRF	fusion	Reciprocal Rank Fusion of TF-IDF + semantic
Hybrid-weighted	fusion	min-max score fusion (α -semantic + $(1-\alpha)$ -lexical)

Passage embeddings are computed once and cached. **Metrics:** Recall@{1,3,5,10}, **MRR**, **nDCG@10**, **MAP**, implemented from their definitions in `src/evaluate.py`.

3.3 Train / validation / test protocol

This is retrieval, so the split is **query-level** (`src/splits.py`):

- **"Train" / index = the corpus** — the lexical models fit their vocabulary + IDF on it and the semantic model encodes it; **no ground-truth labels are used here.**
- The **40 labeled queries** are split, **stratified by domain, seed 42**, into **validation/dev (14, 35%)** and **held-out test (26, 65%)** — written to `data/processed/eval_split.json`.
- No hyper-parameters were tuned on the labeled queries (defaults RRF $k=60$, fusion $\alpha=0.5$), so the split is an **unbiased robustness check**: does the full-set finding survive on queries that influenced no choice? Splitting at the query level (not the chunk level) is what prevents leakage.

4. Results

Retriever	R@1	R@3	R@5	R@10	MRR	nDCG@10	MAP
TF-IDF	0.475	0.725	0.825	0.850	0.618	0.649	0.588
BM25	0.500	0.650	0.825	0.875	0.612	0.647	0.578
AlephBERT (semantic)	0.475	0.700	0.725	0.775	0.579	0.597	0.540
Hybrid (RRF)	0.550	0.750	0.800	0.875	0.668	0.704	0.652
Hybrid (weighted)	0.550	0.725	0.800	0.850	0.653	0.681	0.632

Figures in `results/figures/`: `metrics_bar.png`, `recall_curve.png`, `winloss_heatmap.png`, `embedding_tsne.png`.
Table above = full 40 queries; the **held-out test** numbers below (`results/metrics_by_split.csv`) confirm the ranking

on unseen queries.

Held-out test set (26 queries) — MRR: Hybrid-RRF **0.690** · Hybrid-wtd 0.659 · BM25 0.606 · TF-IDF 0.603 · AlephBERT 0.585. The hybrid lead is *larger* on the held-out test than on the full set, so it is not an artifact of evaluating on the selection data.

Recall@1 by phrasing-gap (H1 test):

gap	n	TF-IDF	AlephBERT	Hybrid-RRF
low	4	0.25	0.50	0.75
medium	11	0.55	0.55	0.64
high	25	0.48	0.44	0.48

Misses (gold not in top-10): TF-IDF 6/40 · AlephBERT 9/40 · Hybrid-RRF 5/40.

Findings.

- **H2 — supported.** The **Hybrid (RRF)** is the best model on every ranking metric (MRR 0.668, nDCG 0.704, MAP 0.652), ties for best Recall@10 (0.875), and misses the fewest queries (5).
- **H1 — not supported (the interesting result).** AlephBERT **alone is the weakest** single model (MRR 0.579, 9 misses) and does **not** beat lexical on high-gap queries (0.44 vs 0.48). A Hebrew sentence encoder derived from a masked LM — not contrastively fine-tuned for retrieval — produces *noisier* similarities than a sharp lexical match in this term-heavy, domain-specific corpus.
- **Why the hybrid still wins.** The semantic model makes *different* errors: it rescues zero-overlap paraphrases lexical cannot (e.g. "מה מגיע לי?" → דמי אבטלה"; gold rank 11→1). Fusing the two keeps lexical precision **and** semantic recall — complementarity, not superiority.

5. Error Analysis

results/error_analysis.md documents 16 worked examples: **4 semantic-wins, 5 lexical-wins, 7 both-fail.**

- **A — semantic wins / lexical fails:** zero/low word-overlap paraphrases ("חייבים להגיע ללשכת התעסוקה כל" → "התניצבות בשירות התעסוקה (הזמן?)"); embeddings bridge the gap.
- **B — lexical wins / semantic fails:** a rare distinctive term gives TF-IDF a sharp signal the embedding diffuses ("כיסא גלגלים ומכשירי שיקום" → exact title; semantic misses).
- **C — both fail:** corpus-coverage gaps / ambiguity ("איך מקבלים קצבת נכות" — the closest page is a Bituach-Leumi stub, drowned by many specific "קביעת נכות בגין... pages).

6. Limitations & Ethics

Small single-annotator eval set (indicative, not tight); six-domain coverage; Hebrew morphology penalizes the lexical models; AlephBERT may carry pre-training social biases. The system is a **research prototype, not legal advice** — a real deployment must show source links and a disclaimer. Full statement in ETHICS.md; data licensing in DATA.md.

7. Conclusion

A complete Hebrew IR pipeline with a controlled comparison of lexical, semantic, and hybrid retrieval for a real social problem. The practical recommendation is the **hybrid**: it combines lexical precision with the semantic model's tolerance of everyday phrasing, and is the only model that beats both baselines. The headline lesson is that an off-the-shelf Hebrew embedding model is **not** a free win for retrieval — but it is a useful *complement*. Future work: a retrieval-tuned model (e.g. E5), Hebrew lemmatization, and a larger graded evaluation set.

8. AI Usage

An AI assistant was used as a **code-writing helper** (boilerplate, docstrings, debugging). All code was reviewed, executed, and is fully understood by the author. See `AI_USAGE.md`.

Personal reflection, failure log, and work log: `REFLECTION.md`.